Model-Directed Attention and the Persistence of Wrong Mental Models

Evidence from firm pricing

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Motivation

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  - (1) DM knows correct model of outcomes; (2) Beliefs about parameters converge to truth.

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  - With fixed wrong model, beliefs stay wrong with infinite data.
  - What if people can revise model? Growing literature studies why people might not switch to a better "mental model," e.g., one provided by a better-informed DM.

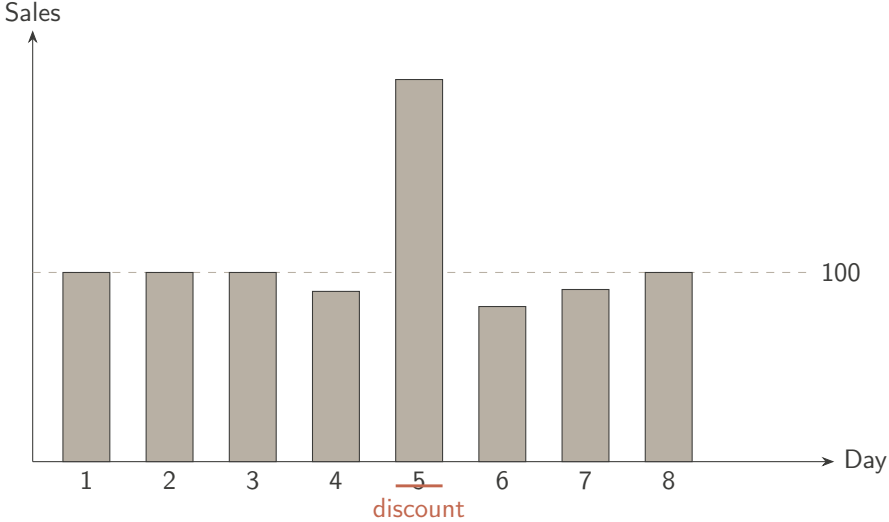
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- This paper tests one such mechanism: **Model-directed attention** (Gagnon-Bartsch et al., 2026)
  - Key idea: Model directs attention to data features it explains.
  - Simple wrong models, silent about what they can't explain, may persist in the face of data: even when disconfirming data are right in front of DM.
  - Encountering a new model may dislodge, if it points to data features neglected under original model.

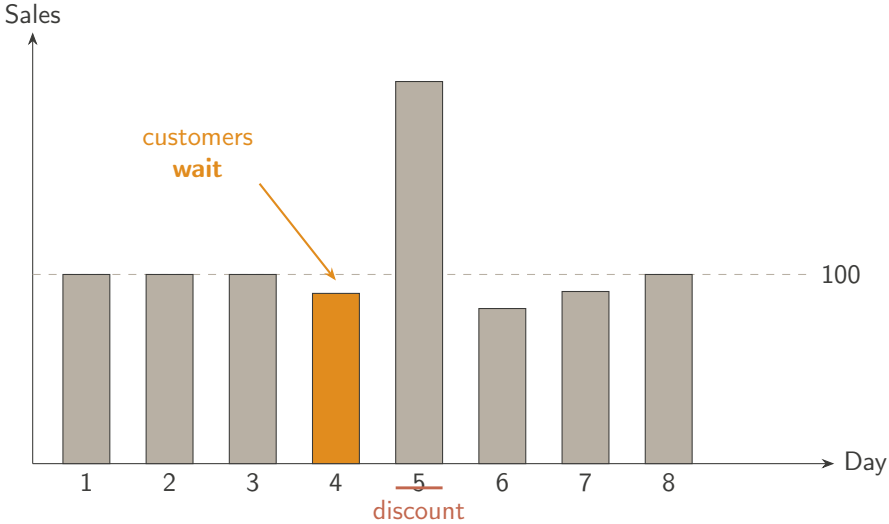
Our focus: Mental model of how price affects consumer behavior

- One of the most fundamental causal relationships studied in economics.
- We hypothesize that some decision makers have a simple, incomplete mental model:
  - When there is a discount, there is an increase in sales, and this reflects increased consumer demand.
- Consider the case of a periodic one-day discount, **advertised ahead of time:** What might data look like?

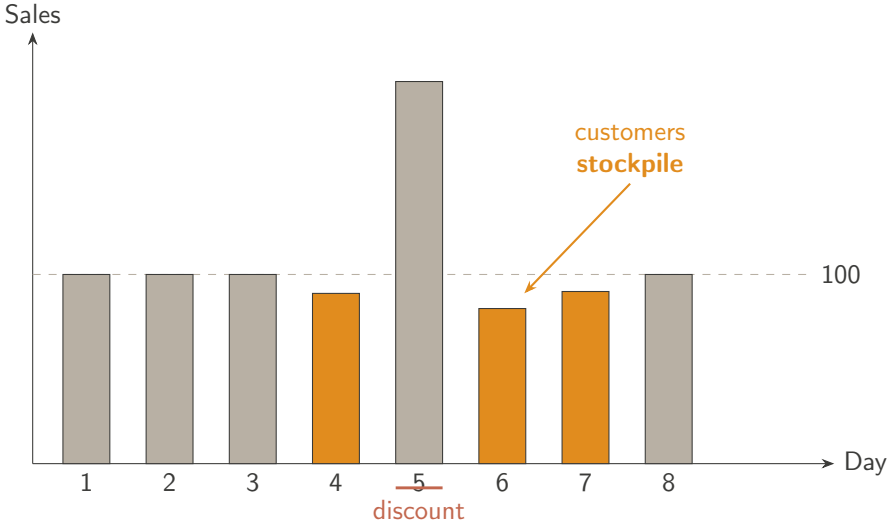
Sales double when price is low, consistent with simple model.



But what about the dip before the discount...



And the dip after the discount. . .



Simple model neglects Intertemporal Substitution (IS)



Read the whole spike as new demand, and the price response looks far larger than it is.

Model-directed attention: How the simple model survives seeing these data

The simple wrong model

$$q_t = D(p_t)$$

One sales value per price
Every regular price day is the same.

The correct model

$$q_t = D(p_t) + \delta(c_t)$$

Sales can vary across regular price days because of *distance to a discount*.

Model-directed attention: How the simple model survives seeing these data

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The model directs attention: is a second regular-price day worth a look?

No. One day calibrates $D(5.9)$; the days are never compared.

Yes. Comparing them reveals the IS displacement $\delta(c_t)$.

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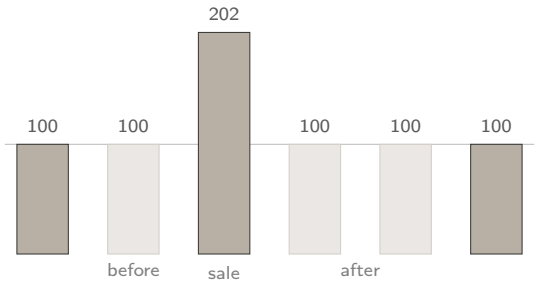
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Simple model is silent about the dip patterns
but attention follows the model so dips are not noticed.

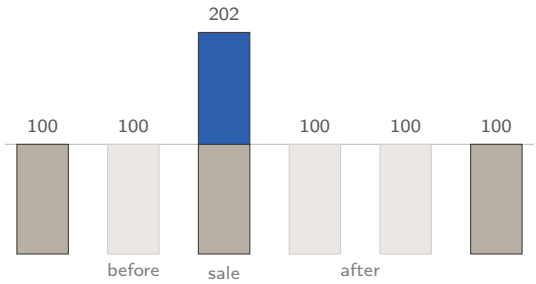
The simple wrong model keeps only two numbers

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Pale bars are **filled in**, not read: one day's 100 stands in for the whole cell

The simple wrong model keeps only two numbers



Our main measure of **IS neglect**: show data like this and ask for predicted daily sales if the price were **always low**

Whole spike read as **new demand**:

- nothing kept **contradicts** it

Predictions

- Models leads to different counterfactual prediction about daily sales if price were always low:
  - Simple model: Sales will be 202.
  - Correct model: Sales will be 165
- Models lead to different beliefs about price elasticity of demand:
  - Simple model: Demand is very elastic
  - Correct model: Demand is less elastic
- Based on different elasticity beliefs, different views on optimal price:
  - Simple model: Deeper price cuts, more frequent (suboptimal, reduces profits).
  - Correct model: Shallower price cuts, less frequent.
  - Note: Although correct model is not as simple, relatively straightforward for DM to use its insights; set price somewhat higher.

Data sources and preview of results

- Measure mental models, decisions of roughly 13,000 gas station managers:
  - Managers have substantial discretion over pricing.
  - Find 20% completely neglect IS when making predictions based on data.
  - Managers neglecting IS are less likely to notice dips in open-text explanations, consistent with model-directed attention.
  - IS-neglect affects elasticity beliefs, charging lower prices.

- Exogenously vary mental models of subjects on Prolific about consumer behavior (Study 1, $N = 549$; Study 2, $N = 896$)
  - Rand. assign explanation by previous participant with simple wrong or correct model.
  - Assigning wrong model leads to inattention to dips and IS neglect.
  - Attention prompt pointing to dips reduce IS neglect, showing causality the other way.
  - A second model can dislodge initial model more effectively than data or pure attention prompt, if it widens attention; can lead to the truth even if it is also incomplete or wrong.

Implications

- This is not a paper about gas stations, instead an illustration of a source of persistent wrong beliefs in the face of data.
  - Tip of the iceberg: Simple wrong models, or "blind spot models" are likely pervasive across almost all decision environments.
  - We show how these can survive in seemingly unforgiving environments: Disconfirming data arrives exogenously.

- The same mechanism has implications for social learning.
  - Directly sharing models may be more effective in correcting beliefs than just providing data, or pure attention prompts.
  - Findings on the benefits of sharing incomplete/wrong mode: Could explain why some organizations encourage idea-sharing even by inexperienced or ill-informed employees.

Contribution to econ literature (1/2)

- **Persistence of wrong models** (theory)
  - Existing explanations: self-confirming data, wrong model has better payoffs, wrong model has lower complexity costs, correct model outside the consideration set

    Hong, Stein & Yu 2007; Ortoleva 2012; Cho & Kasa 2015; Schwartzstein & Sunderam 2021; Heidhues, Kőszegi & Strack 2021; Fudenberg & Lanzani 2023; Gagnon-Bartsch, Rabin & Schwartzstein 2026
  - Our findings are not explained by these: data are exogenous, the wrong model is costly, and attention prompts neither reduce complexity nor propose the correct model

- **Biased beliefs despite knowing the correct model** (rational inattention, sparsity)
  - Field: wrong beliefs persist despite large costs, and the correct model is accessible and acted on by others; online treatments are orthogonal to these mechanisms

    Sims 2003; Gabaix 2014, 2019; Maćkowiak, Matějka & Wiederholt 2023

- **Experiments on mental models and narratives**
  - We measure the impact of a mental model on attention to specific data features, manipulate attention directly, and test asymmetric model switching

    Kendall & Charles 2022; Barron & Fries 2024; Esponda, Vespa & Yuksel 2024; Ambuehl & Thysen 2025

Contribution to econ literature (2/2)

- **Behavioral firms and IO**
  - Managers' wrong models of consumers lead to lower prices and profits

    Hendel & Nevo 2006; Heidhues & Kőszegi 2018; DellaVigna & Gentzkow 2019; Strulov-Shlain 2023; Han et al. 2025
  - Some discounts may reflect managers' misperception of consumers, not dynamic price discrimination

- **Social learning and communication of models**
  - Sharing models can be an important source of learning

    Hong & Page 2004; Cai & Szeidl 2018; Schwartzstein & Sunderam 2021; Montiel Olea et al. 2022
  - Even an incomplete or wrong model can lead others to the truth by redirecting attention

Related literature from psychology

- **What we are doing shapes what we see and remember**
  - Task focuses attention: counting basketball passes, viewers miss a gorilla walking through the scene
  - Role shapes recall: the same story read as a burglar or a homebuyer yields different remembered details

  ⇒ **Our contribution:** attention is driven by a *belief about how the world works*, as opposed to task or role, and shapes beliefs and economic decisions.

- **Biased testing of causal relationships in data**
  - Covariation neglect: judging a relationship mostly from cases where both features appear

  ⇒ **Our contribution:** not a general difficulty with using data to test hypotheses; the *same* data are read differently depending on the *model* a person is given

  - Positive test strategy: people generate only data their hypothesis predicts, so the wrong rule is never refuted

  ⇒ **Our contribution:** refuting data are *exogenous*, complete and in plain view, rather than generated by the person's own choices

Mental models of consumer behavior

Main measure is based on a data table shown to decision makers

- Subjects are told:
  - These "data" are generated by survey designers, using a set of rules.
  - Can be thought of as reflecting online detergent sales.
  - Everything determining sales is in the table.
  - Dataset is representative.

- These instructions are intended to help subjects understand that the dataset is deterministic, self-contained without omitted factors, not a fluke generated by a particular sample.

Data in table form (as managers typically see data)

Day	Retailer A's price	Competitor B's price	Retailer A's sales volume	Retailer A's profit
Day 1	5.9	5.9	100	160
Day 2	5.9	5.9	100	160
Day 3	5.9	5.9	100	160
Day 4	5.9	5.9	90	144
Day 5	5.2	5.9	202	182
Day 6	5.9	5.9	82	131
Day 7	5.9	5.9	91	146
Day 8	5.9	5.9	100	160
Day 9	5.9	5.9	100	160
Day 10	5.9	5.9	90	144
Day 11	5.2	5.9	202	182
Day 12	5.9	5.9	82	131
Day 13	5.9	5.9	91	146
Day 14	5.9	5.9	100	160

Retailer A's regular price is **5.9**.

The two weeks contain **two discount days**, days 5 and 11, at a price of **5.2**.

Competitor B charges 5.9 throughout.

Data show three key patterns

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- Purple shows sales dip due to waiting; green shows increase due to new customers+IS; yellow shows dip due to stockpiling

Multiple choice question to elicit models, asked on same screen as data

"If retailer A always charges 5.2 per kg, what will the daily sales volume be?"

192	**165**	100	**202**
Partial-IS model	Correct model	IS-only model	Simple wrong model

- Answers are proxies for models.

192 and 100 correspond to two other wrong models that could survive

Partial-IS model

Customers *wait* for a discount,
but never *stock up* during it.

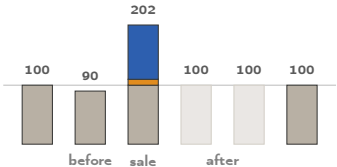
IS-only model

A discount changes *when* people buy,
never *how much* they buy.

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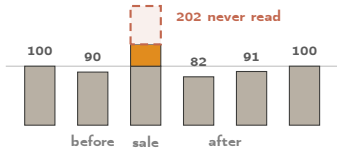
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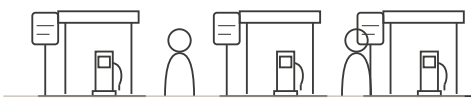


IS-only model

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shifted from nearby days new demand from the lower price filled in, not read



Field evidence

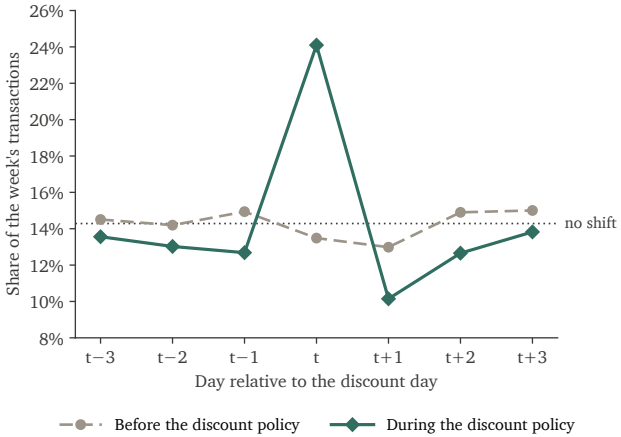
Partner with a company that operates 20,000+ gas stations

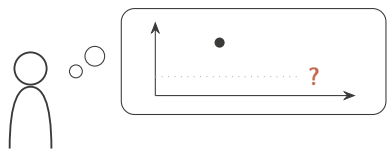
- Data-rich, high-stakes environment:
  - Station managers have substantial discretion over setting fuel prices
  - Incentivized based on station profits
  - They have access to high frequency data on sales and profits
  - Substantial experience, median is 7 years

- Roughly 13,000 respond to our online survey that presents the detergent dataset and associated questions.
- We can match to data on actual pricing decisions, profits.

This is a market where consumers exhibit substantial IS

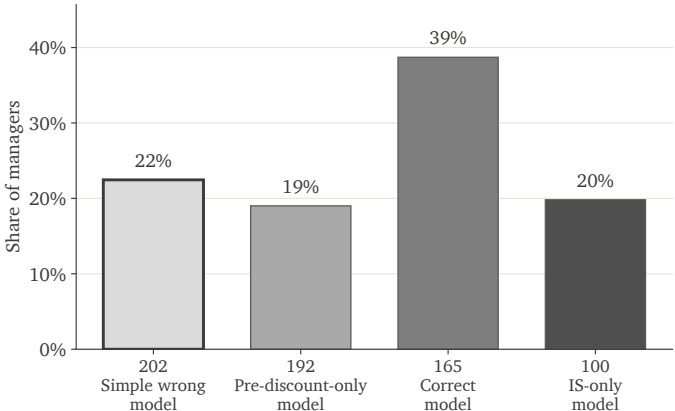
- We look at the 24% of stations who start offering regular weekly discounts
- See strong shift of regular customers to discount day



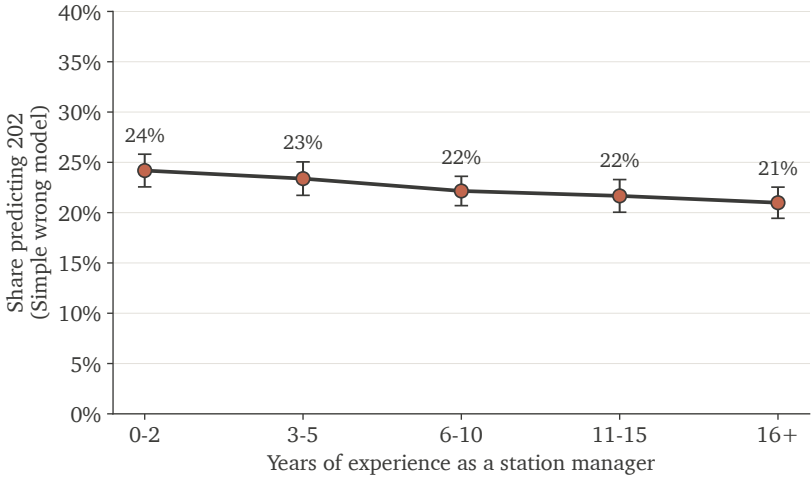


Mental models of managers

One manager in five sees no intertemporal substitution



Experience does not remove it



Then we asked them to explain the same table

"Looking at days 4–7 and days 10–13,
why do you think sales show this pattern
before and after the days when price is 5.2?"

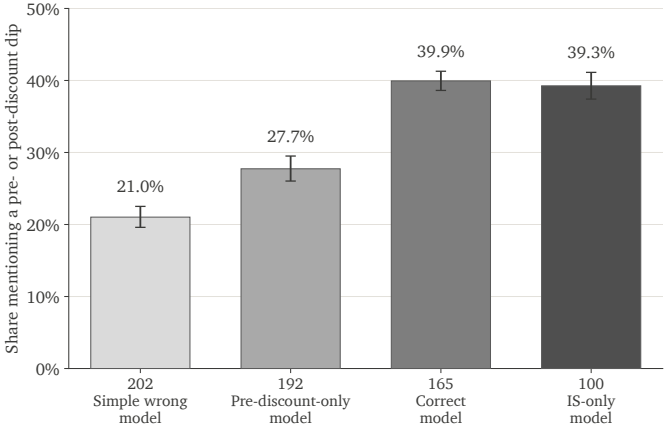
Open text, straight after the table. 13,407 answers.

Answers differ in whether they mention dip patterns

"The price reduction stimulated customers' purchasing desire, leading to increased sales."

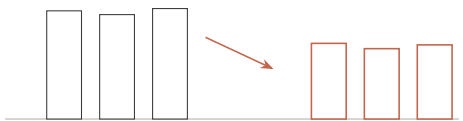
"Customer traffic **decreased the day before** and **two days after** the promotion, as purchases were concentrated on the discount day."

Managers predicting 202 are less likely to notice the dips



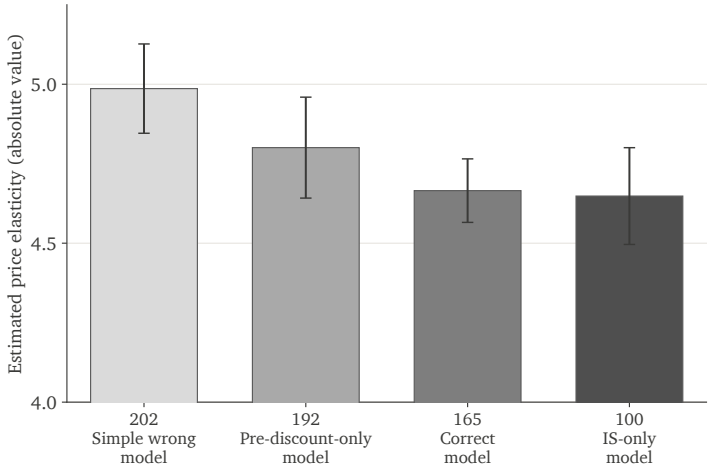
192 versus 202: +6.7 percentage points ($p < 0.001$).

Noticing inferred from the full explanation; $N = 13{,}407$.

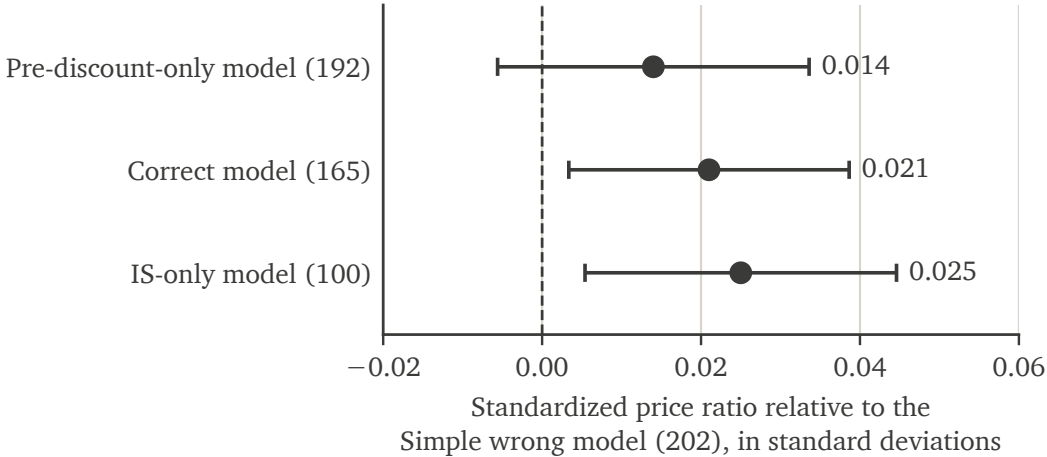


Field consequences

They think fuel demand is more elastic



And they charge lower prices



Estimating the profit cost

1. Estimate demand in the one region with daily prices and sales.

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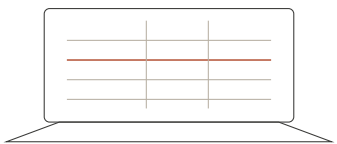
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4. Calculate the differences in profits.

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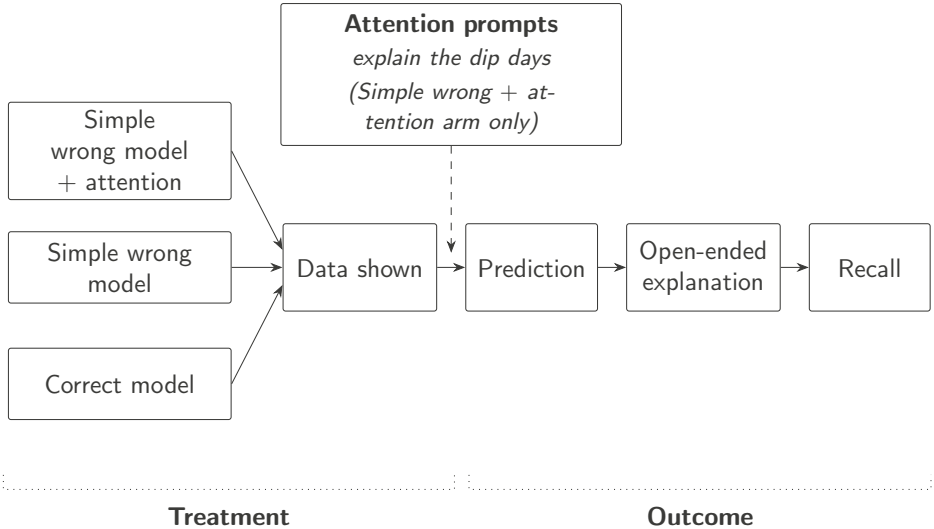
About 3.3% of daily fuel profit.



Experiment

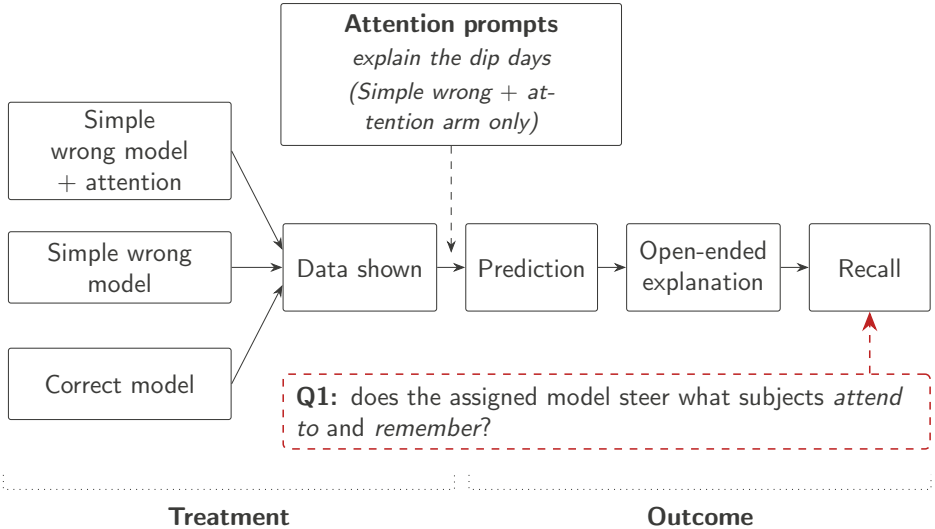
Design of Study 1

$N = 549$, **pre-registered**



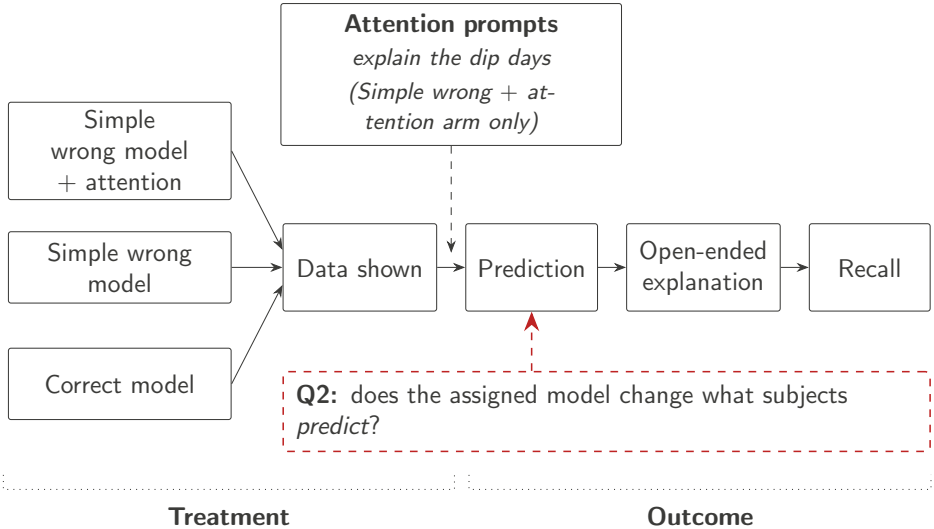
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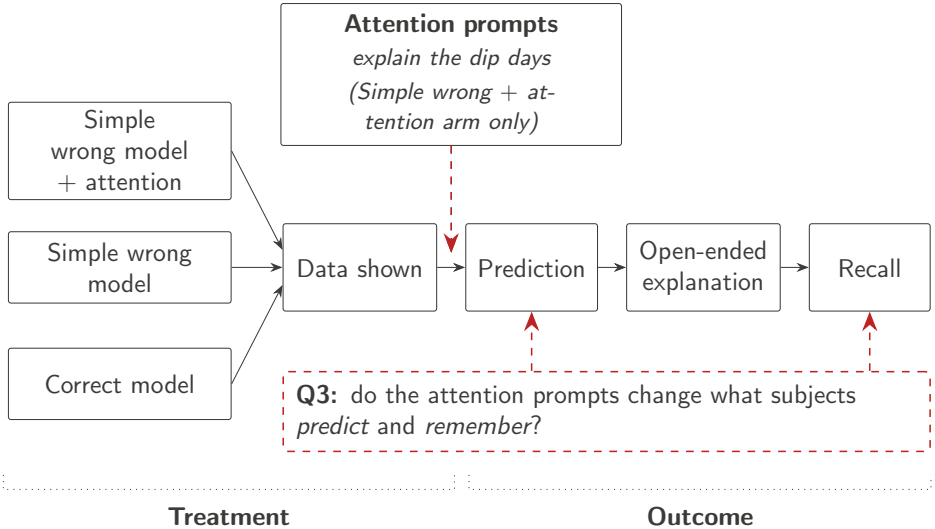
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The treatments are model descriptions from previous participants predicting 202, or 165, respectively

Simple wrong model

"When the price is low, sales are greater than when the price is high. This has to do with the rule of supply and demand. When the prices are low demand increases."

Explains only the **spike** pattern

↑ Intended to induce the **simple (wrong)** model

Correct model

"...sales surge on that discount day but drop on the surrounding days... because customers postpone purchases to wait for the discount and then have already stocked up afterward."

Explains both the **spike** and the **dip** patterns

↑ Intended to induce the **correct** model

What we measure

- **A prediction.** Sales at a permanent 5.2, incentivized.
- **An explanation.** What rule generated the data, incentivized.
- **Confidence.** How much belief they place on their own answer.
- **Recall.** Every one of the seven days, incentivized.

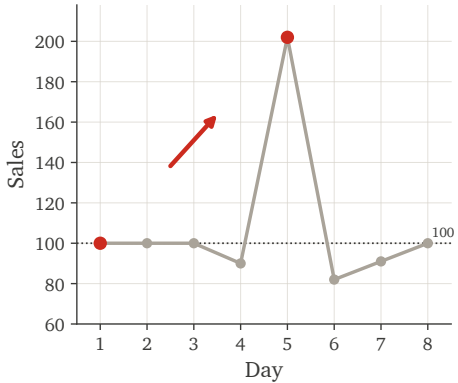
Did model assignment move attention?

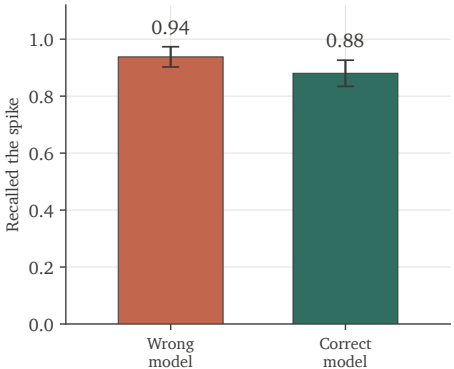
The typical recalled week, under each model

	Data shown		Median recalled sales	
Day	Price	Sales	*Wrong model*	*Correct model*
1	5.9	100	100	100
2	5.9	100	100	100
3	5.9	100	100	100
4	5.9	**90**	100	92
5	5.2	202	202	202
6	5.9	**82**	100	90
7	5.9	**91**	100	93.5

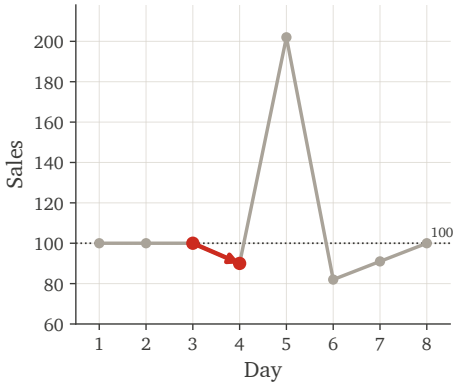
Median recalled sales among those who answered.

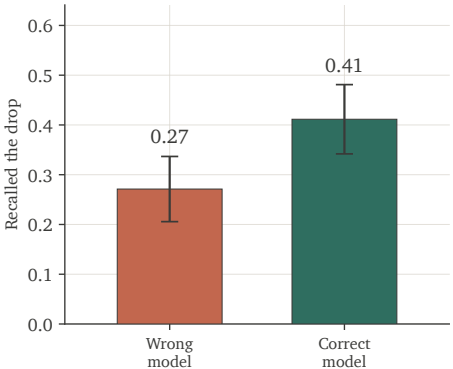
Everyone remembers the day the model explains



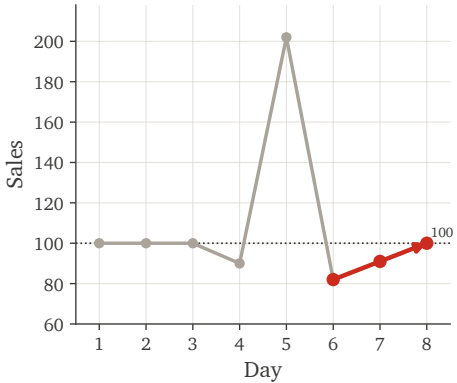


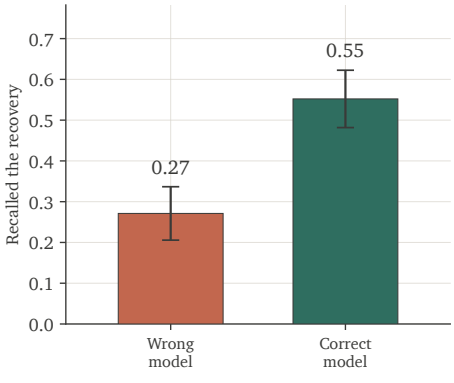
Fewer remember the drop before it





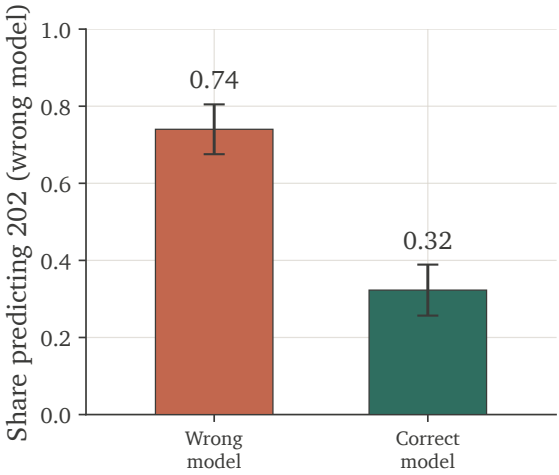
And fewer still the recovery after it

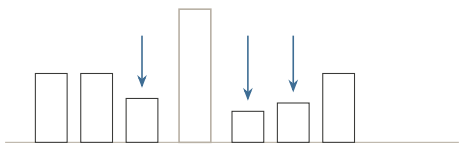




Did model assignment move prediction?

A peer's wrong model more than doubles the wrong answer





Attention intervention

The attention intervention is two questions about the data

With the table in view, and *before* the prediction:

What do you think explains the pattern in Retailer A's sales from Day 3 to Day 4, and from Day 9 to Day 10?

What do you think explains the pattern in Retailer A's sales from Day 6 to Day 8, and from Day 12 to Day 14?

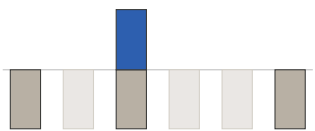
Directing attention without explanation can dislodge the wrong model

A prompt to **compare the two days before the discount** adds no data and no new model: it only redirects attention.

Directing attention without explanation can dislodge the wrong model

A prompt to **compare the two days before the discount** adds no data and no new model: it only redirects attention.

Simple wrong model

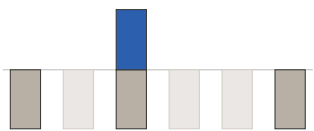


regular-price days are never compared

Directing attention without explanation can dislodge the wrong model

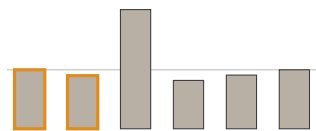
A prompt to **compare the two days before the discount** adds no data and no new model: it only redirects attention.

Simple wrong model



regular-price days are never compared

After the prompt



100 *and* 90 at the same price
⇒ **simple model refuted**

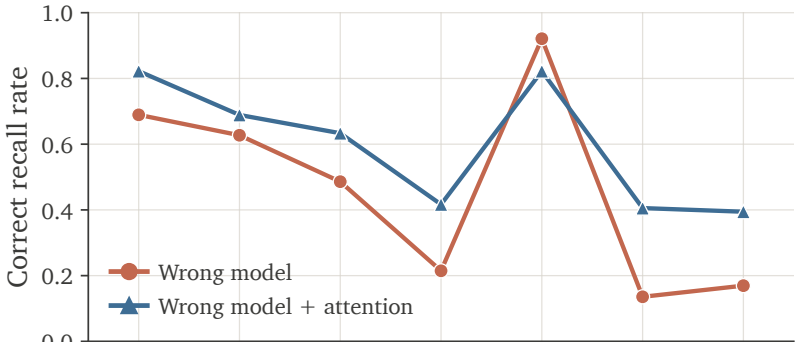
Directing attention without explanation can dislodge the wrong model

A prompt to **compare the two days before the discount** adds no data and no new model: it only redirects attention.



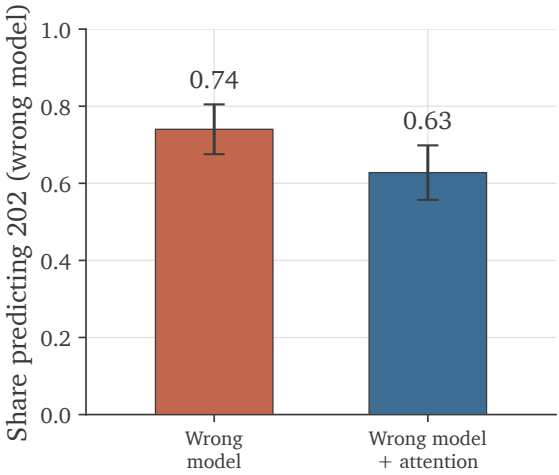
What changes is not the data, but what is **noticed**.

Pointing at the dips successfully redirected attention



Sales	100	100	100	90	202	82	91
Price	5.9	5.9	5.9	5.9	5.2	5.9	5.9

And moved people off the wrong model



Additional points about attention prompt

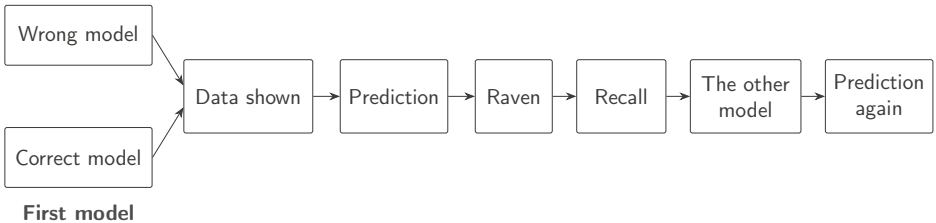
- Manipulation check showed we successfully increased attention to dips.
- Treatment effect shows that inattention to dips is helping sustain the incorrect model.
- Why not a bigger correction?
  - Attention prompt could have been more heavy handed, potentially increasing impact.
  - Noticing dips is a pre-requisite for figuring out the correct model, but doesn't mean someone automatically figures it out.
  - More powerful way to correct is combination of attention with explanation, i.e., directly communicating the correct model?



The second model

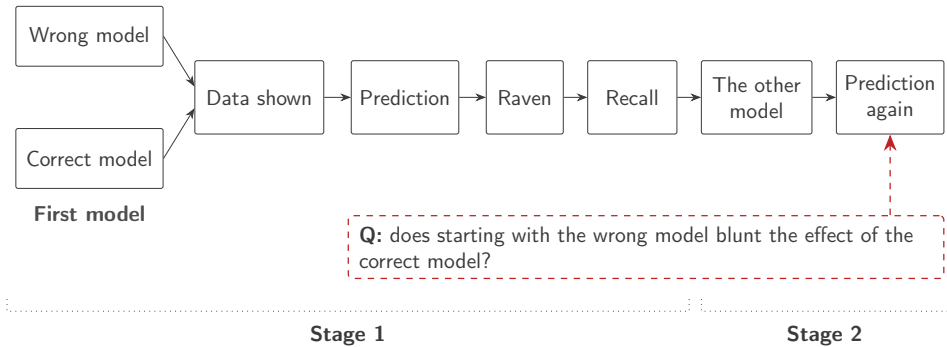
Design of Study 2

$N = 896$

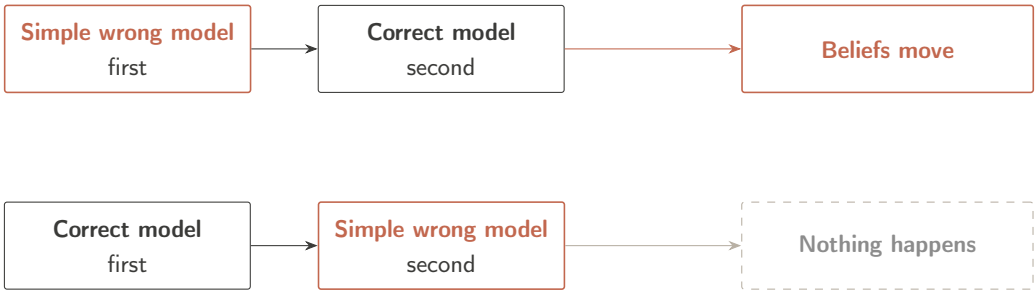


Design of Study 2

$N = 896$



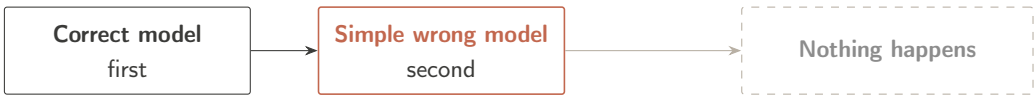
Model-directed attention implies an asymmetric effect of order



Model-directed attention implies an asymmetric effect of order



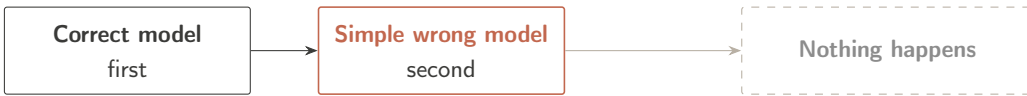
Entertaining the correct model points to the dips that refute the simple wrong model.



Model-directed attention implies an asymmetric effect of order



Entertaining the correct model points to the dips that refute the simple wrong model.

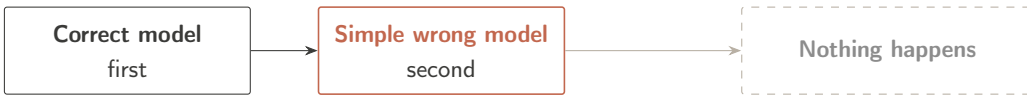


The dips were already noticed. The wrong model is refuted by what has already been noticed.

Model-directed attention implies an asymmetric effect of order



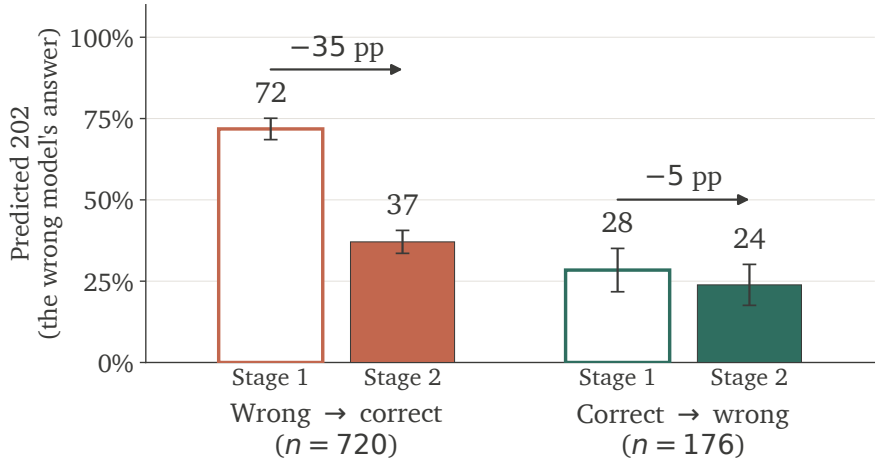
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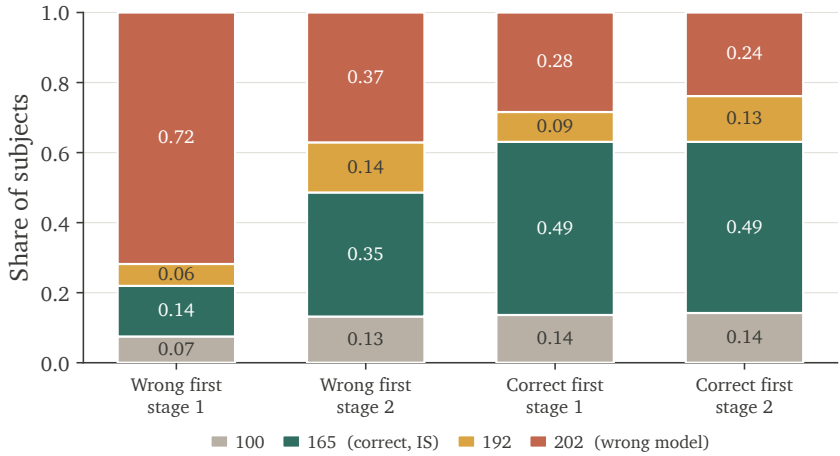
An effect in one order, and none in the other.

Correct model can dislodge wrong; wrong does not dislodge correct



Within person, across the two stages. Re-showing the data changes nothing (−35 vs −34 pp).

Wrong-first subjects improve, but do not entirely catch up



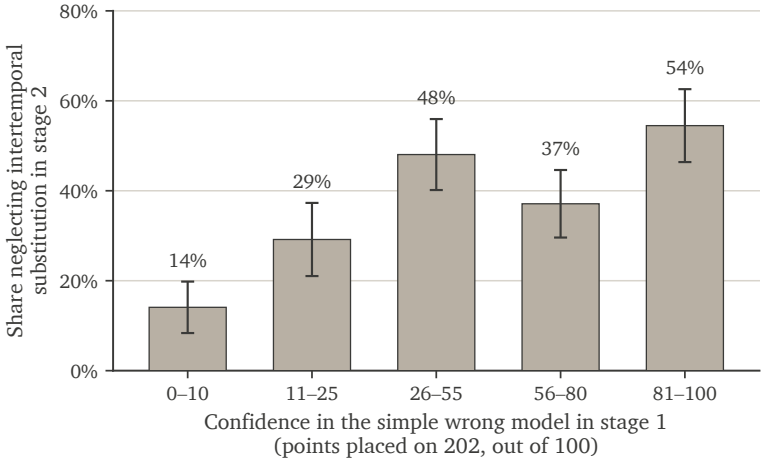
Also illustrates that imperfectly expressed/understood model can still lead to the truth

- Our correct model treatment uses the typical statement of pilot subjects who predict 165, but this is incomplete.
  - The statement mentions dips and spike as outcome of IS, but doesn't explicitly mention new demand.

- Because it widens attentional focus relative to simple wrong model, however, it can lead to the truth.

- We see that as a second model, it increases the fraction giving correct answer (165) by 21 ppt., increases the IS-only answer (100) by only 6 ppt.

What predicts subjects not “entertaining” the second model? Confidence in initial prediction.



Conclusion

Takeaways

- **Overly-simple mental models can persist in the face of data due to model-directed attention.**
  - Wrong beliefs may survive even in seemingly unforgiving data rich environments.
- **Encountering attention-expanding models can dislodge wrong models**
  - May not fully close the gap.
  - Benefit of even incomplete models shows value of idea sharing.
- **Broad implications: Models of this form may arise in many important decision environments.**
  - The cases of Moneyball in baseball, three-pointers in basketball.
  - Rental car company neglect of data showing that customers like renting older cars for low prices.
  - Physicians neglecting predictors of heart attacks.
  - Investors ignoring analyst affiliation.
  - Beliefs about effectiveness of cold remedies and other medicines.

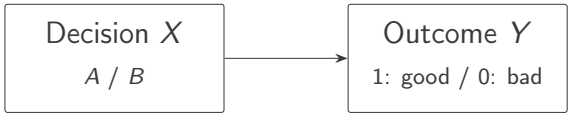
Takeaways

- **Neglect of IS is important in its own right.**
  - Overestimate demand elasticity, charge prices that are too low, and earn lower profits.

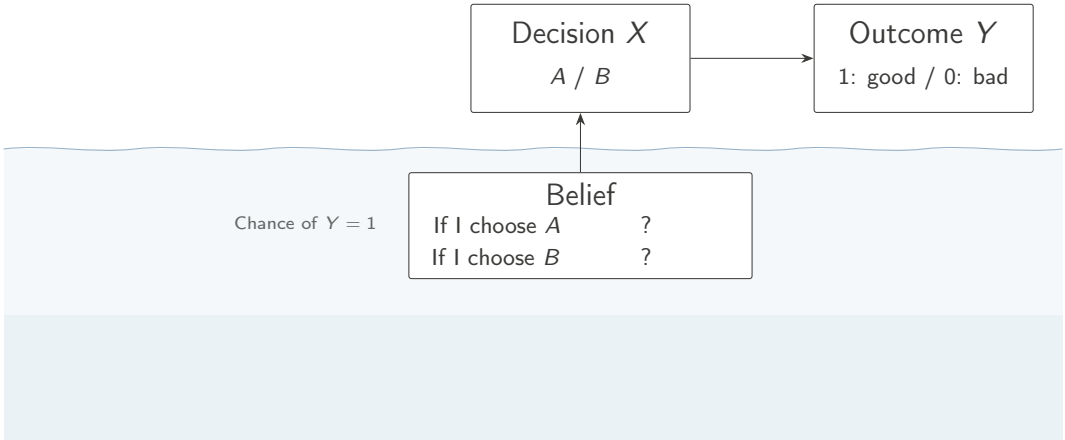
Thank you!

Appendix: Research agenda

Mental models and economic outcomes



Mental models and economic outcomes



Mental models and economic outcomes



Decision X

A / B

Outcome Y

1: good / 0: bad

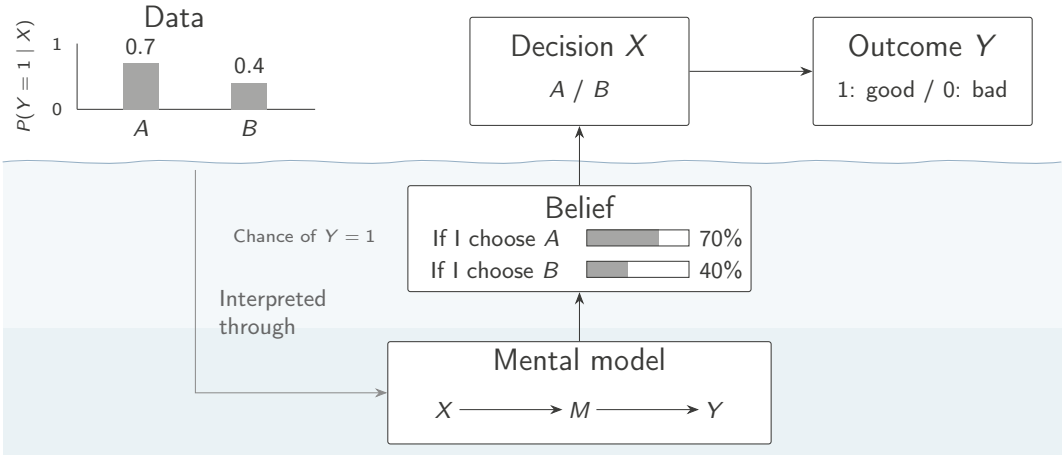
Chance of $Y = 1$

Belief

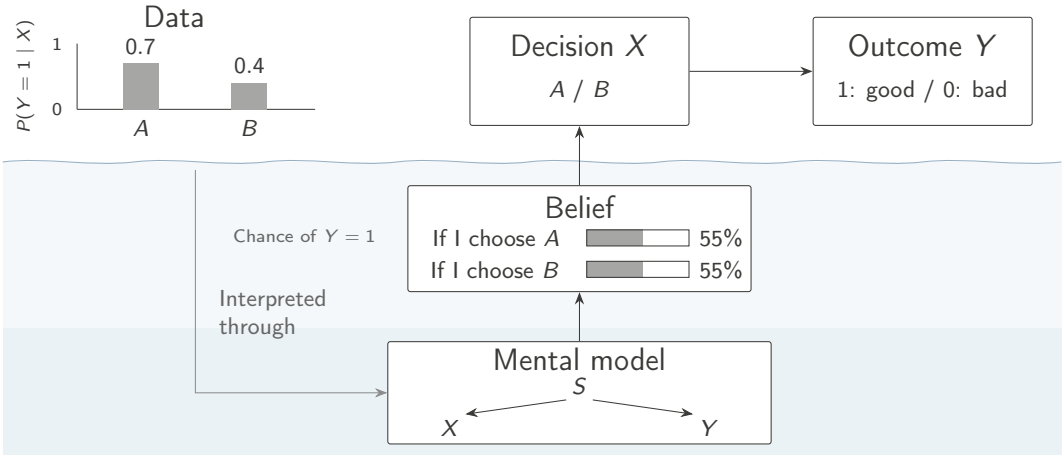
If I choose A ?

If I choose B ?

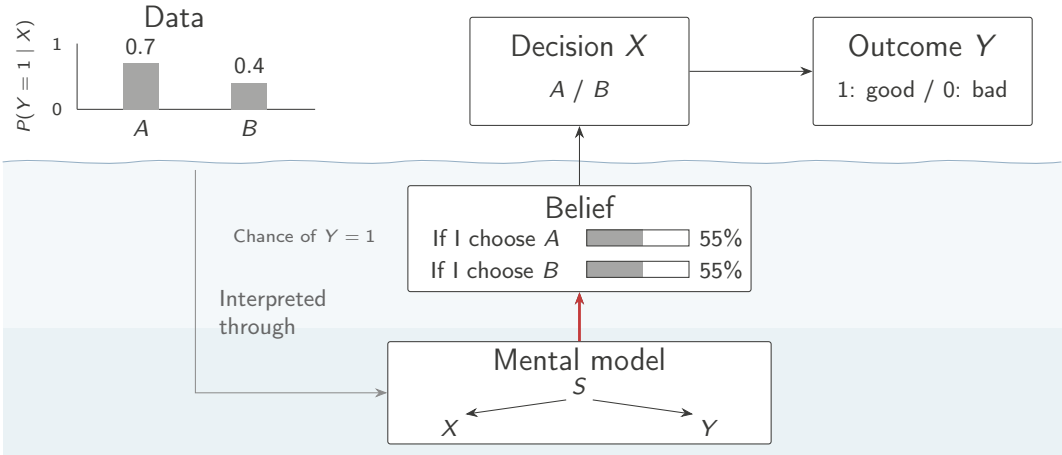
Mental models and economic outcomes



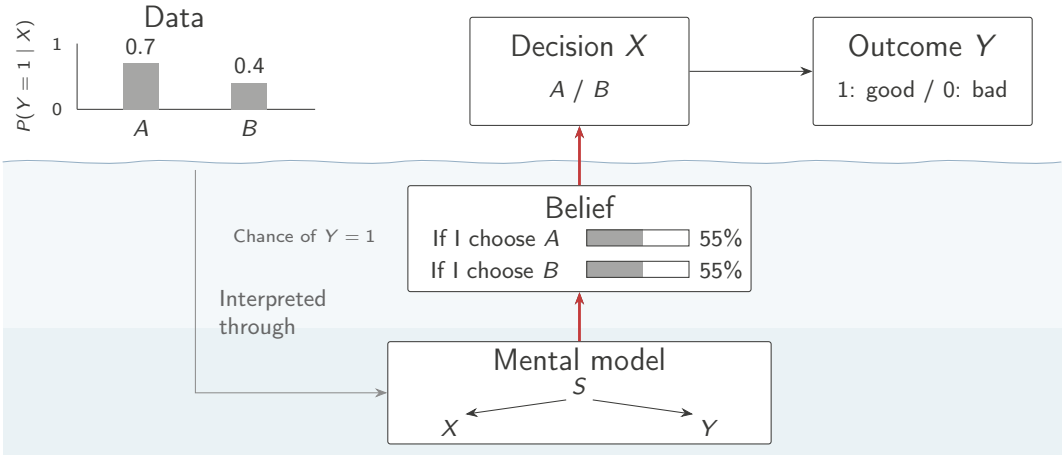
Mental models and economic outcomes



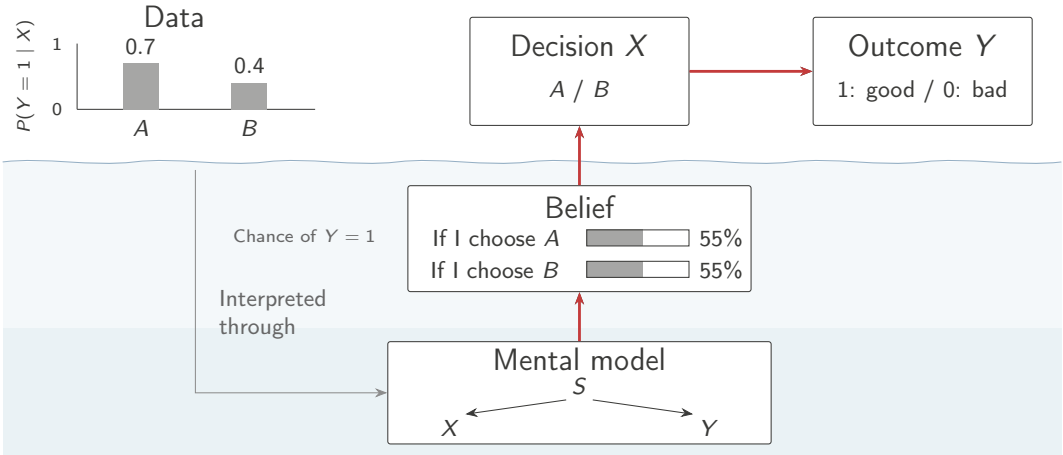
Mental models and economic outcomes



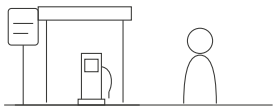
Mental models and economic outcomes



Mental models and economic outcomes



Field relevance + mechanism depth



Field evidence

Trace outcomes

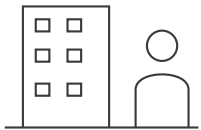
+



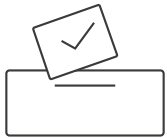
Controlled experiments

Identify mechanisms

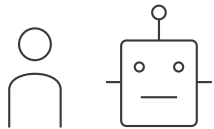
One framework, three domains



Managerial decision-making

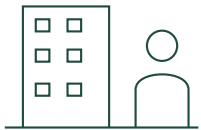


Public choice



Human–AI interaction

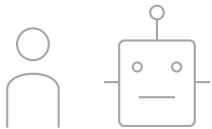
One framework, three domains



Managerial decision-making



Public choice



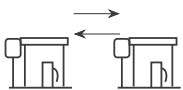
Human–AI interaction

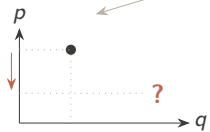
Managers: models, decisions and outcomes

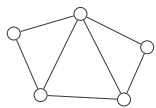
Manager survey
mental models and beliefs
13,407 station managers

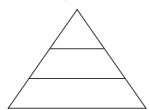
Administrative data
station prices, sales and profits
16,000 stations, 2019–2024

matched



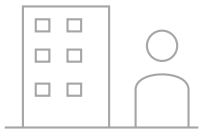






Competitors	Consumers	Networks	Hierarchy
R&R at *AER*	JMP	in progress	pre-registered

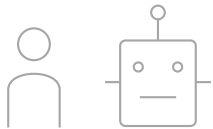
One framework, three domains



Managerial decision-making

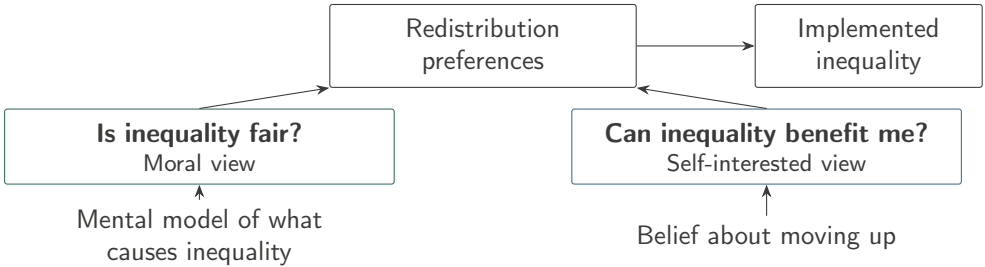


Public choice

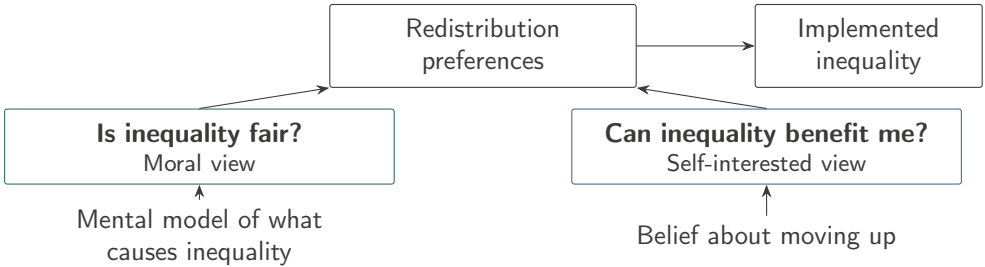


Human–AI interaction

Two beliefs behind redistribution



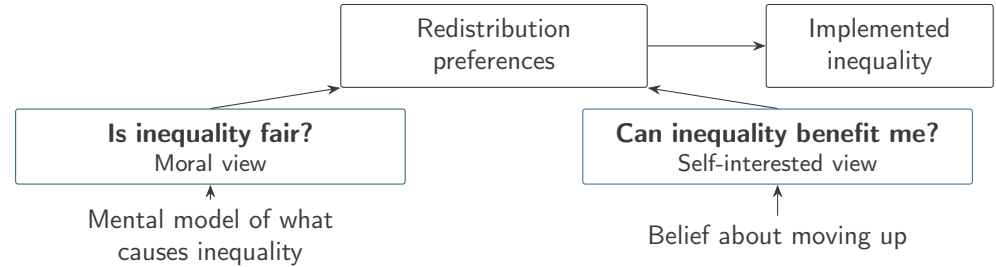
Two beliefs behind redistribution



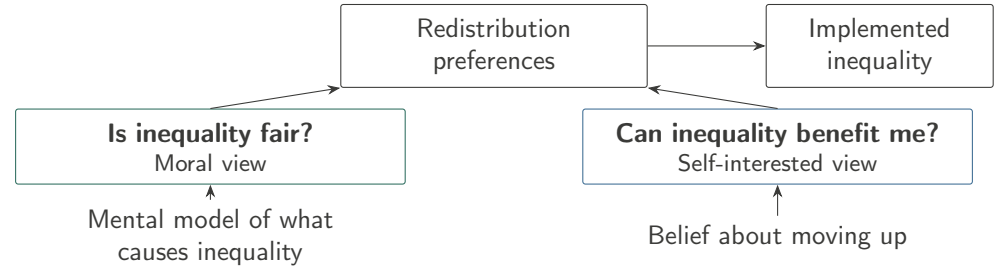
Fairness and attribution bias

Unequal opportunity $\xrightarrow{?}$ Effort $\longrightarrow$ Income

Two beliefs behind redistribution



Two beliefs behind redistribution



Fairness and attribution bias

Unequal opportunity → (?) Effort → Income

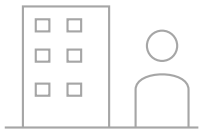
Dreaming Rich

Inequality → Mobility belief → Inequality acceptance

Voice and fairness (planned)

Worker's mental model → Narrative → Spectator's mental model

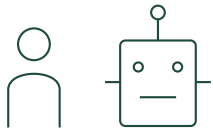
One framework, three domains



Managerial decision-making



Public choice



Human–AI interaction

Three levels of AI integration

	AI's role	Decision authority	Research question
1	**Adviser**	AI advice → Human decision	**Who asks AI for advice, when, and how?** Strategic interaction
2	**Delegate**	Human instructions → AI action	**Can AI act as the person would, given their instructions?** Alignment
3	**Judge**	AI moral judgment → Final decision	**Do AI moral principles differ from ours?**

Other papers in agenda order

Managers

1. Minds, Models and Markets

2. Networks and mental models

3. Own versus Others' Information

Inequality

4. Dreaming Rich

5. Fairness and Attribution Bias

6. Voice and Fairness

The agentic economy

7. One Advisor, Many Advisees

8. Ex Ante Alignment

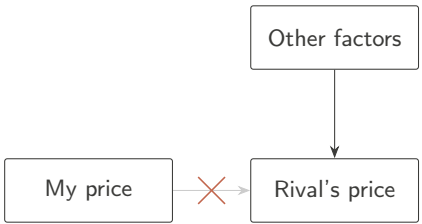
9. AI distributive choices

Select a paper to jump to its slides

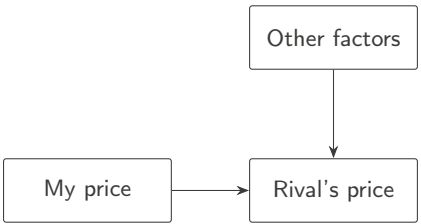
Does my price affect my rival's price?

Minds, Models and Markets · R&R at the AER

Response omitted

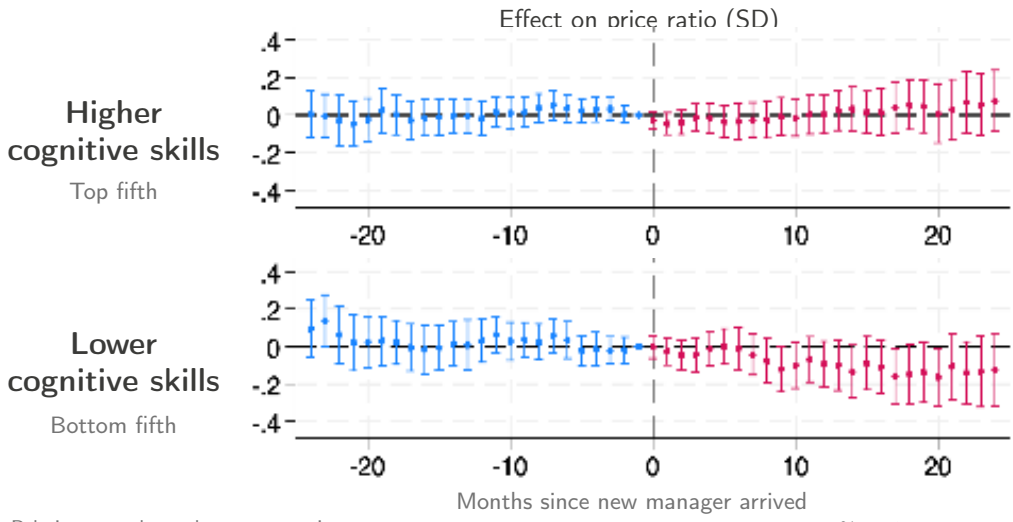


Response included



With Yi Han and David Huffman

Prices diverge after a new manager arrives



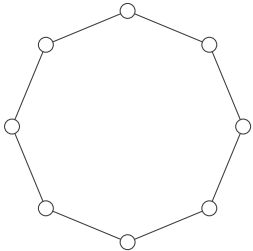
Relative to unchanged-manager stations

95% confidence intervals

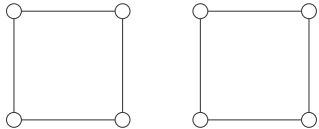
Same links, different fragmentation

in progress

Cedar: connected



Maple: fragmented



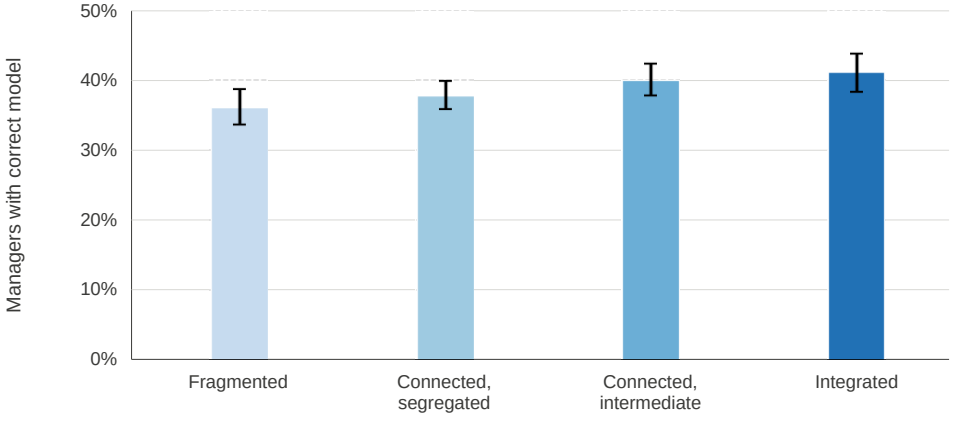
8 managers · 8 links · 2 links each

Which networks sustain correct mental models?

Illustration; fictional names

Integrated networks have more correct models

in progress

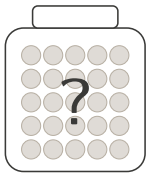


Intertemporal substitution · Branch means · Intervals as reported

Own information versus a colleague's

pre-registered

The game: 3 rounds, a new jar each round



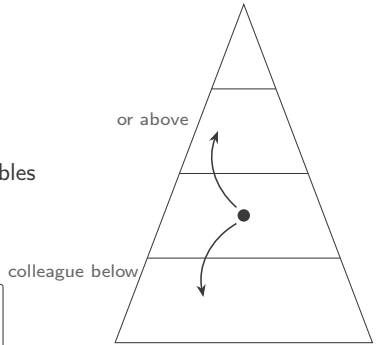
Your 5 marbles



A colleague's 5 marbles



Guess: how many dark marbles are in the jar?



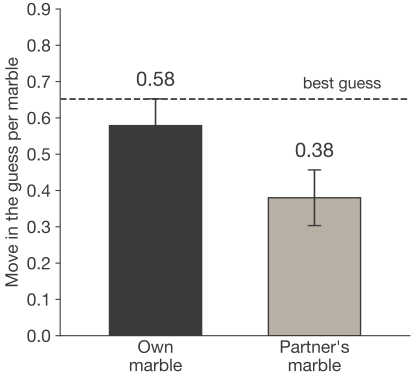
Own over colleague's?

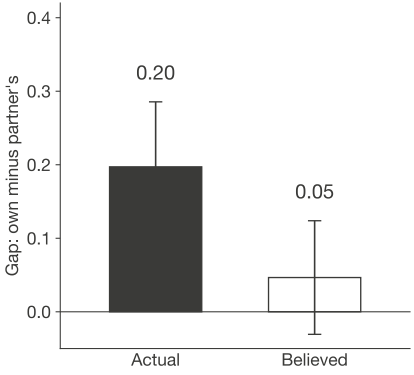
Depends on the colleague's level?

Does the top know?

Own versus Others' Information · task: Conlon et al. (2026)

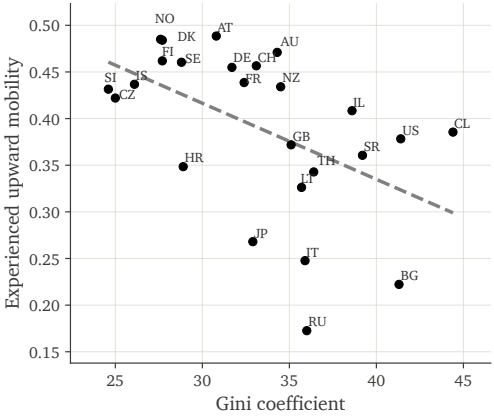
Online pilot ($N = 238$): a partner's marbles count a third less

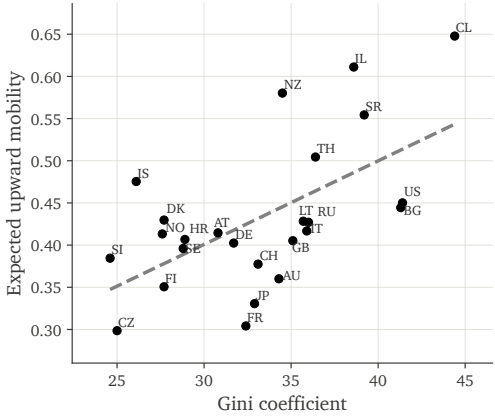




Own versus Others' Information · Field study pre-registered

Dreaming Rich: less mobility, higher hopes





With Julia Baumann · Working paper · 25 countries

High inequality inflates hopes of reaching the top

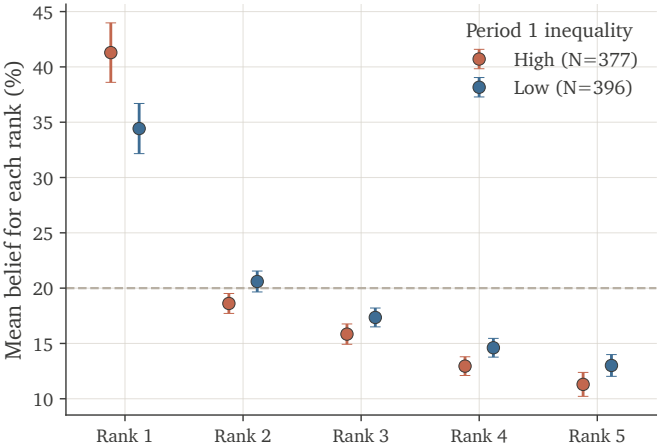
Rich



Rich: 20% on average

Rich vs each other
High: \$10 vs \$0
Low: \$2.40 vs \$1.90

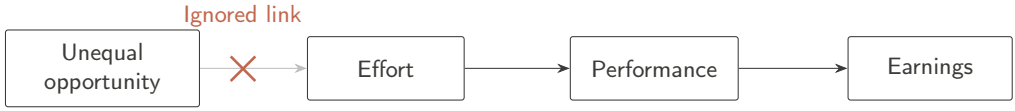
same total income



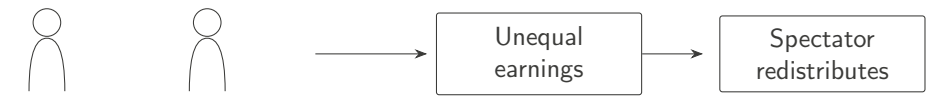
Dreaming Rich · Working paper · Baseline, $N = 773$

When opportunity is mistaken for merit

Fairness and Attribution Bias · R&R, *Management Science*



Random piece rates



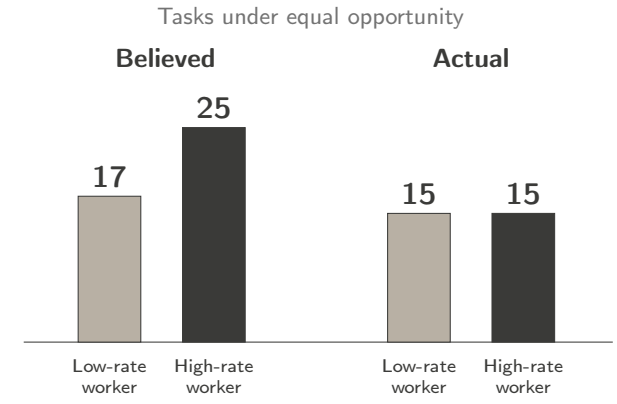
With Cappelen, Nielsen and Tungodden

Unequal opportunities are read as unequal merit

Fairness and Attribution Bias · R&R, *Management Science*

81%

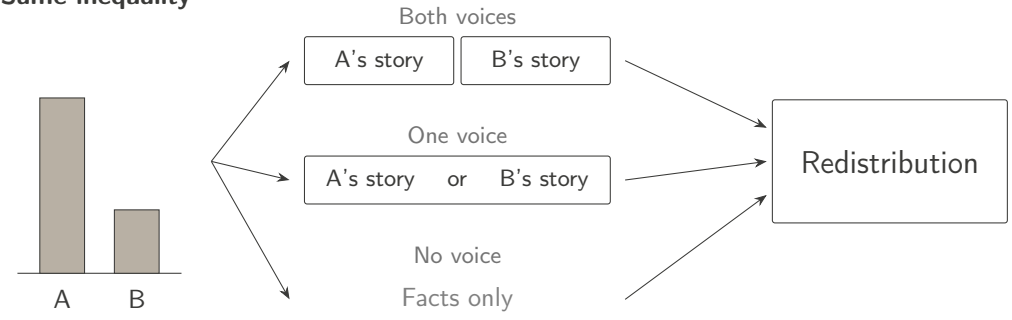
leave the advantaged worker ahead



Voice and fairness: whose argument wins?

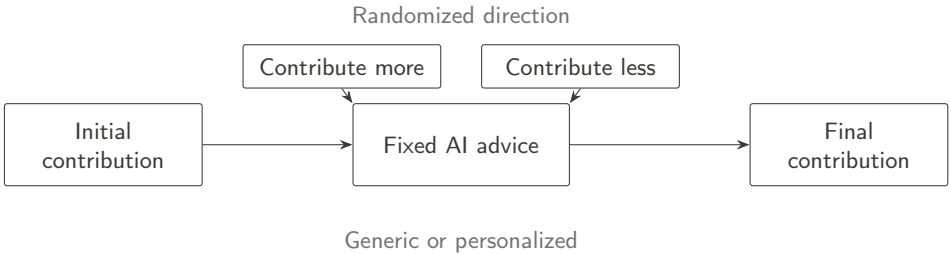
With Cappelen and Tungodden · Planned

Same inequality



Workers' own narratives; facts held fixed

How does AI advice change a decision?

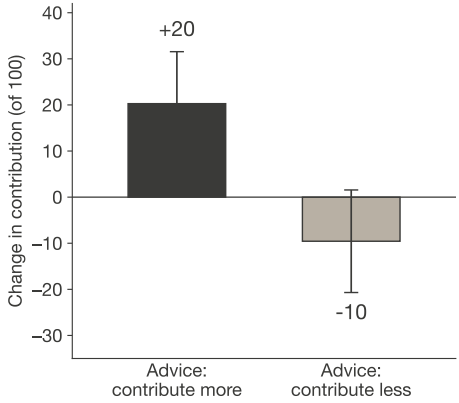


Public goods game · 100 points · $N = 41$

with Davies, Jackson, Ye and Yuan

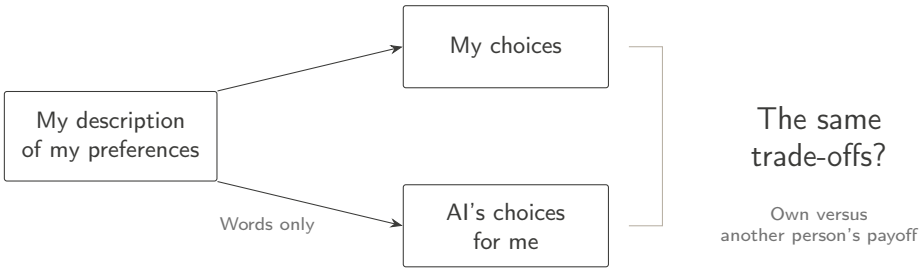
Advice shifts contributions in both directions

Pilot, $N = 41$



95% confidence intervals

Can an AI choose as I would?



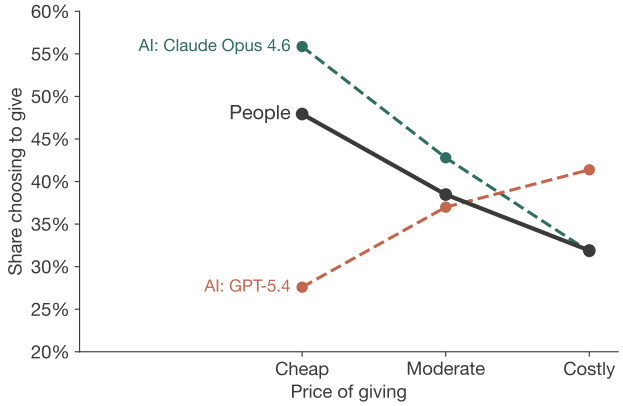
58 people · 23 allocation choices · 10 AI models

with Jackson, Mei and Yuan

An AI delegate can reverse the price response

Delegate · *Ex Ante Alignment*

Pilot, $N = 58$



Would you redistribute, at a cost?

won the lottery

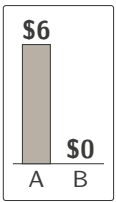


A
$6

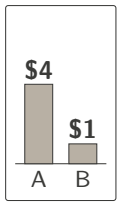


B
$0

Each $1 you give B costs A $2



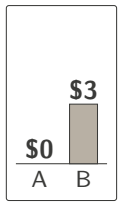
total $6



total $5



total $4



total $3

?

Most people pay for equality

won the lottery



Chose the equal split, $2 and $2

People

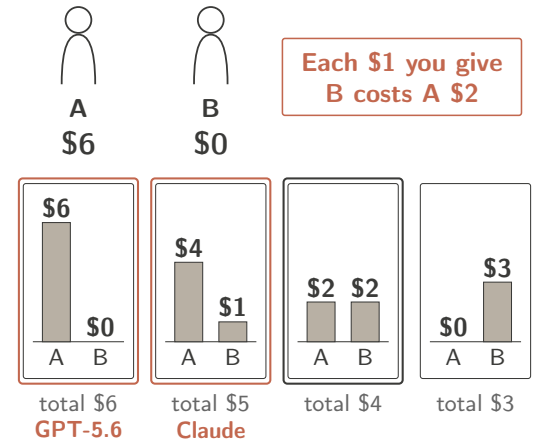


People: Almås, Cappelen & Tungodden (2020)

Two AI agents choose less equality

Decision maker · *Distributive choices* Pilot, with Erkut

won the lottery



Chose the equal split, $2 and $2

People

US **58%**

Norway **76%**

AI, deciding for a user

Claude Opus 5 **19%**

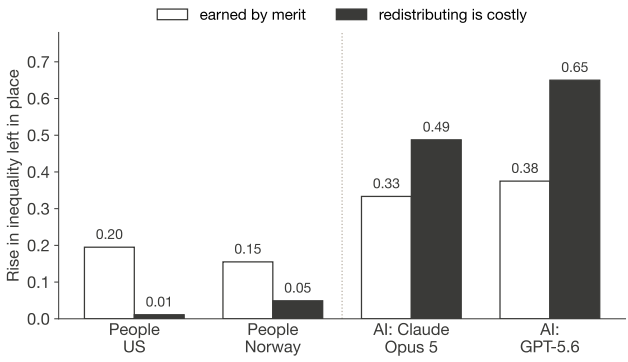
GPT-5.6 **25%**

16 draws per model

People: Almås, Cappelen & Tungodden (2020)

Cost outweighs merit for these two AI agents

Decision maker · *Distributive choices* Pilot



Relative to lottery, no redistribution cost

with Erkut · People: Almås, Cappelen & Tungodden (2020) · AI: 16 draws per cell